

Table 5. Information about datasets and their splits.

| Dataset         | Train  | Validation | Test (Public/Private) |
|-----------------|--------|------------|-----------------------|
| IOI             | 10000  | 10000      | 1000/1000             |
| MCQA            | 110    | 50         | 50/50                 |
| Arithmetic (+)  | 34400  | 4920       | 1000/1000             |
| Arithmetic (−)  | 17400  | 2484       | 1000/1000             |
| ARC (Easy)      | 2251   | 570        | 1188/1188             |
| ARC (Challenge) | 1119   | 299        | 586/586               |
| RAVEL           | 100000 | 16000      | 1000                  |

Table 6. An IOI example and its 8 associated counterfactuals.

| Prompt / Counterfactual                    | Name A | Name B | Name C | Text                                                                                     | Correct Completion |
|--------------------------------------------|--------|--------|--------|------------------------------------------------------------------------------------------|--------------------|
| Original Prompt                            | Nick   | John   | Nick   | <i>After Nick and John spent some time at the car dealership, Nick offered a nail to</i> | John               |
| ABC                                        | Nick   | John   | Bob    | <i>After Nick and John spent some time at the car dealership, Bob offered a nail to</i>  | N/A                |
| Random Names                               | Max    | Fred   | Max    | <i>After Max and Fred spent some time at the car dealership, Max offered a nail to</i>   | Fred               |
| IO ↔ S1 Flip                               | John   | Nick   | Nick   | <i>After John and Nick spent some time at the car dealership, Nick offered a nail to</i> | John               |
| IO ↔ S2 Flip                               | Nick   | John   | John   | <i>After Nick and John spent some time at the car dealership, John offered a nail to</i> | Nick               |
| Random Names + IO ↔ S1 Flip                | Fred   | Max    | Max    | <i>After Fred and Max spent some time at the car dealership, Max offered a nail to</i>   | Fred               |
| Random Names + IO ↔ S2 Flip                | Max    | Fred   | Fred   | <i>After Max and Fred spent some time at the car dealership, Fred offered a nail to</i>  | Max                |
| IO ↔ S1 Flip + IO ↔ S2 Flip                | John   | Nick   | John   | <i>After John and Nick spent some time at the car dealership, John offered a nail to</i> | Nick               |
| Random Names + IO ↔ S1 Flip + IO ↔ S2 Flip | Fred   | Max    | Fred   | <i>After Fred and Max spent some time at the car dealership, Fred offered a nail to</i>  | Max                |

## C.2. Arithmetic

We list the templates used to format the arithmetic queries in Table 7. We consider four text-based prompts and two Arabic numeral-based prompts. The prompts are modified from [Stolfo et al. \(2023\)](#). An example instance and its associated counterfactuals are in Table 8. Subtraction queries are constrained to cases with positive results to maintain single-token answers when possible. Of the four models we investigate, two (Llama and GPT-2) tokenize numeric answers as single tokens and two (Qwen and Gemma) tokenize numbers into their respective digits (meaning that correct answers will often be more than one token in length). In the latter case, we generate the oracle number of tokens corresponding to the number of digits in the correct answer and then check for exact-match correctness.

**Counterfactuals.** We create counterfactuals by adjusting the operands in ways that will affect not only the correct answer, but also the addition or subtraction process (see Table 8). In our experiments, we primarily use the *random operands* counterfactual for baseline comparison, but we provide the additional counterfactuals for further analysis.

## C.3. Multiple-choice question answering (MCQA)

We expand the dataset of [Wiegrefe et al. \(2025\)](#), itself based on [Norlund et al. \(2021\)](#), by adding 102 additional instances from [Paik et al. \(2021\)](#) whose object group is “0”, indicating that participants agreed on a prototypical color for that

Table 7. Prompt templates for single-operator two-operand arithmetic operations.

| Template | Addition                                       | Subtraction                                  |
|----------|------------------------------------------------|----------------------------------------------|
| 1        | Q: How much is $n_1$ plus $n_2$ ? A:           | Q: How much is $n_1$ minus $n_2$ ? A:        |
| 2        | Q: What is $n_1$ plus $n_2$ ? A:               | Q: What is $n_1$ minus $n_2$ ? A:            |
| 3        | Q: What is the result of $n_1$ plus $n_2$ ? A: | Q: What is the result of $n_1$ minus $n_2$ ? |
| 3        | The sum of $n_1$ and $n_2$ is:                 | The difference between $n_1$ and $n_2$ is:   |
| 5        | $n_1 + n_2 =$                                  | $n_1 - n_2 =$                                |
| 6        | $n_1 + n_2 =$                                  | $n_1 - n_2 =$                                |

Table 8. An Arithmetic example and its 7 associated counterfactuals.

| Prompt / Counterfactual           | Text                            | Correct Completion |
|-----------------------------------|---------------------------------|--------------------|
| Original Prompt                   | <i>The sum of 27 and 64 is:</i> | 91                 |
| Random Operands                   | <i>The sum of 42 and 29 is:</i> | 71                 |
| Different ones digit in operand 1 | <i>The sum of 24 and 64 is:</i> | 88                 |
| Different ones digit in operand 2 | <i>The sum of 27 and 61 is:</i> | 88                 |
| Different tens digit in operand 1 | <i>The sum of 47 and 64 is:</i> | 111                |
| Different tens digit in operand 2 | <i>The sum of 27 and 44 is:</i> | 71                 |
| Different ones digit carry value  | <i>The sum of 21 and 60 is:</i> | 81                 |
| Different tens digit carry value  | <i>The sum of 77 and 64 is:</i> | 141                |

object—for example, that bananas are yellow. We randomly sample 3 incorrect colors from a set of 11 and pair them with the correct answer choice in a random position to create each instance. By design, each answer in this task is a single token (e.g., A, B, C, D).

In the paper, we report on 4-choice MCQA, as this is a standard number of choices and allows comparison with the ARC dataset. We also create versions of the dataset with 2, 3, 5, 6, 7, 8, 9, and 10 answer choices, in order to allow for future investigation into how mechanisms change as the result of having fewer or more choices.

**Counterfactuals.** We create two semantic counterfactuals and three format counterfactuals for each instance (and four combinations of these, resulting in 9 counterfactuals total). Semantic perturbations involve replacing the noun in the question (such as “banana”) with a different noun from another instance in the same split, or the correct color of the noun mentioned in the question (such as “yellow”) with another color (such as “brown”). The latter changes the correct answer; the former does not.

Format perturbations do not change the correct color itself, but do change the symbol that represents that color, and therefore, change the correct answer. We follow a similar design to Lieberum et al. (2023a). We change the position of the correct answer, the symbols representing the answer choices (i.e., 1/2/3/4 instead of A/B/C/D), or the letters (i.e., the randomly selected sequence E/Z/F/L instead of A/B/C/D). See Table 9 for an example dataset instance and its associated counterfactuals.

#### C.4. AI2 Reasoning Challenge (ARC)

By design, each answer in this task is a single token (e.g., A, B, C, D), making the prompt format similar to MCQA (Appendix C.3).

**Counterfactuals.** The counterfactual types and generation process are identical to the MCQA counterfactual generation process (described in Appendix C.3). Due to the varying content of each ARC prompt and the lack of a token-level template between prompts, we include only the format-based counterfactuals, omitting *semantic* counterfactuals such as “Noun” and “Color” from MCQA. For an example prompt and its counterfactuals, see Table 10.

Table 9. An MCQA example and its 9 associated counterfactuals.

| Prompt / Counterfactual         | Text                                                                                                                     | Correct Completion |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------|
| Original Prompt                 | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>A. gray\B. black\C. white\D. pink\Answer:</i>     | D                  |
| Noun                            | <i>Question: A banana is pink. What color is a banana?</i><br><i>A. gray\B. black\C. white\D. pink\Answer:</i>           | D                  |
| Color                           | <i>Question: Salmon meat is yellow. What color is salmon meat?</i><br><i>A. gray\B. black\C. white\D. yellow\Answer:</i> | D                  |
| Noun+Color                      | <i>Question: A banana is yellow. What color is a banana?</i><br><i>A. gray\B. black\C. white\D. yellow\Answer:</i>       | D                  |
| Answer Position                 | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>A. gray\B. black\C. pink\D. white\Answer:</i>     | C                  |
| Symbol                          | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>1. gray\2. black\3. white\4. pink\Answer:</i>     | 4                  |
| Random Letter                   | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>E. gray\Z. black\F. white\L. pink\Answer:</i>     | L                  |
| Answer Position + Random Letter | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>E. gray\Z. black\F. pink\L. white\Answer:</i>     | F                  |
| Answer Position + Symbol        | <i>Question: Salmon meat is pink. What color is salmon meat?</i><br><i>1. gray\2. black\3. pink\4. white\Answer:</i>     | 3                  |
| Answer Position + Color         | <i>Question: Salmon meat is yellow. What color is salmon meat?</i><br><i>A. gray\B. black\C. yellow\D. white\Answer:</i> | C                  |